

BiasLens — Project Documentation & Mentor Presentation

Comprehensive Overview, Problem Statement, System Architecture, Features & Client-Side NLP Engine Logic

1. Project Overview & Vision

BiasLens is an open-source, media-literacy web application designed to empower readers by revealing how different news outlets frame the exact same real-world event. Through a side-by-side article reader, real-time natural language processing (NLP), interactive 3D stacked benchmark cards, and Explainable AI (XAI) tooltips, BiasLens makes media bias, loaded phrasing, and editorial framing transparent, measurable, and easily understandable.

Project Name:	BiasLens (NLP Media Literacy Analyzer)
Target Audience:	Students, Journalists, Researchers, General Public seeking balanced media consumption
Core Execution:	100% Client-side in-browser NLP execution (Zero backend latency, total user privacy)
Key Innovation:	Interactive 3D article card stack, dual-panel reader, XAI word neutralizer popovers

2. Problem Statement & Why We Need BiasLens

Modern media consumers face unprecedented challenges due to algorithmic echo chambers, partisan media polarization, and high-intensity sensationalism:

- **Information Echo Chambers & Confirmation Bias:** Most readers stick to single news outlets that reinforce their pre-existing beliefs, remaining unaware of alternative viewpoints or loaded adjectives.
- **Sensationalism & Clickbait Framing:** Outlets frequently use high-emotion, sensationalist words (e.g. 'catastrophic', 'slammed', 'draconian') to drive clicks rather than report objective facts.
- **Selective Omission & Framing Divergence:** Two outlets covering the same legislative vote or economic policy will characterize key actors and institutions in opposite ways (e.g. 'Historic Win to Protect Workers' vs. 'Draconian Scheme Throttle Growth').
- **Lack of Real-Time Media Literacy Tools:** Consumers lack easy-to-use, visual tools to compare news coverage side-by-side in real time and see *why* specific words carry bias.

■ Why 100% Client-Side Architecture?

By building our NLP engine natively in browser JavaScript, BiasLens requires zero backend server costs, operates instantly with no network latency, and ensures total user privacy — as article text is never sent to external third-party tracking servers.

3. Core Objectives

- Side-by-Side Comparison:** Render news articles on identical events in parallel to contrast headlines, tone, and framing.
- Real-Time Language Scanning:** Automatically highlight sensationalism, loaded political terms, and subjective opinion markers.
- Explainable AI (XAI):** Allow users to click any highlighted word to view feature attribution details and see neutral alternative phrasing.
- Quantitative Bias Metrics:** Display visual charts for political stance spectrums, sensationalism indices, factual objectivity ratings, and entity alignment matrices.
- Custom Article Analysis:** Provide a dedicated floating glass panel where users can paste any two news articles for instant analysis.

4. Technology Stack

Layer	Technology / Tool	Role & Purpose
Core Frontend	React 18 + Vite	Component-based architecture, fast state management, modular components, instant HMR
Styling & Design	Vanilla CSS (Glassmorphism)	Custom dark design system (`#080a10`), backdrop blur (`blur(24px)`), grid mesh overlays, 3D card stack
Icons	Lucide React	Clean, modern SVG icons for UI indicators, badges, and action buttons
NLP Computing	JavaScript NLP Engine (`nlpEngine.js`)	Custom regex tokenization, lexicon dictionaries, sentiment analysis, headline skew, entity alignment
Graphics & Assets	AI Generated Artwork	Custom editorial artwork thumbnails for AI Act, Climate Policy, and Global Economy cards

5. Logic of the NLP Engine (Easy Explanation)

The NLP engine in BiasLens evaluates news text through a 6-stage algorithmic pipeline:

Stage 1: Tokenization & Normalization

The article body text is split into individual sentences using sentence boundary detection regex (`/[^\!?\+][.!?]+\b/g`). Each sentence is then broken down into cleaned lowercase word tokens while retaining punctuation for smooth rendering.

Stage 2: Lexicon & Dictionary Matching

Each word token is checked against curated linguistic dictionaries:

- Sensational Words:** Words designed to provoke strong emotional arousal (e.g., *devastating, catastrophic, slammed, bloodbath, chaos, crushed*).
- Loaded Words:** Words carrying heavy political or ideological framing (e.g., *draconian, bureaucrats, cartel, regime, radical, syndicate*).

- **Opinion Markers:** Words asserting subjective editorial judgment (e.g., *clearly, obviously, tragically, foolishly, shockingly*).

Stage 3: Sentiment & Objectivity Scoring

Words are evaluated against positive lexicon (*historic, triumph, landmark, pioneering*) and negative lexicon (*collapse, crisis, bleak, ruinous*) to compute sentence-level and article-level net sentiment scores.

```
Sensationalism Index = min(100, (Sensational Words / Total Words) * 1200)
Loaded Language Index = min(100, (Loaded Words / Total Words) * 1400)
Objectivity Rating = max(0, 100 - (0.4 * Sensationalism + 0.4 * Loaded + 0.2 * Subjectivity))
```

Stage 4: Headline vs. Body Skew Analysis

BiasLens checks if the headline uses high-arousal clickbait terms while the underlying body text remains neutral. If headline sensationalism exceeds body sensationalism significantly, a **Headline Skew Alert** is raised.

Stage 5: Entity Framing Divergence Matrix

The engine identifies key actors (e.g. 'Government & Policymakers', 'AI Act', 'Climate Policy', 'Economy & Central Bank'). It extracts all sentences mentioning these entity keywords and computes the localized sentiment score to determine whether each outlet frames that specific entity as **Favorable**, **Critical**, or **Neutral**.

Stage 6: Explainable AI (XAI) & Neutralizer Suggestions

When a user clicks on any flagged word in the text, an XAI popover displays feature attribution reasoning and automatically maps the word to a neutral substitute (e.g. replacing 'slammed' with 'criticized', 'draconian' with 'strict', 'bureaucrats' with 'officials').

6. Key UI Modules & Application Features

1. **3D Interactive Article Card Stack:** Fanned horizontal cards in the hero section displaying benchmark articles with artwork, category badges, and bias rating pills.
2. **Sticky Glassmorphic Navbar:** Features brand identity (Bias in red, Lens in white), live model indicator badge, quick reset button, and custom article trigger button.
3. **Dual Article Reader Panel:** Side-by-side article reader with filter pills (All, Sensational, Loaded, Subjective) and interactive word highlights.
4. **XAI Popover Modal:** Interactive popover explaining why a highlighted token was marked and providing a neutral rewrite.
5. **Comparative Analytics Dashboard:** Includes Political Stance Spectrum bar gauge, loaded language progress meters, and objectivity ratings.
6. **Entity & Keyword Matrix:** Matrix comparing how specific entities are framed across outlets.
7. **Custom Article Floating Panel:** Modal dialog allowing users to input custom article pairs for instant NLP analysis.

7. Summary & Presentation Talking Points for Mentor

When presenting BiasLens to your mentor, highlight these 4 key takeaways:

1. **Solves a Real Social Problem:** Improves media literacy by helping users identify emotional manipulation and biased framing in news.
2. **Zero-Cost & Private Architecture:** 100% client-side execution means instant speed, zero server bills, and total privacy for readers.
3. **Modern Premium Aesthetics:** Implements glassmorphism, 3D article card stacks, and curated color tokens for a stunning user experience.
4. **Explainable & Actionable:** Doesn't just call an article 'biased'; it pinpoints exact words, explains **why**, and suggests neutral alternatives.